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[Exploratory Data Analysis]
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Analysis

Variable definition

DATA filtered to only show cells with more than 30 observations.
Loans are correctly priced... perfectly balanced... as all things should be.
However, default rates are very high for the riskier segments. So high that the high interest rates might not compensate the high default rate.

PD VS Int. rate & Grade

PD VS Int. rate & Grade

	Grade						
Int Rate (bi..	A	B	C	D	E	F	G
5	3.28%	7.56%	3.70%	2.68%	4.41%	16.00%	12.50%
10		8.10%	13.96%	14.29%	4.69%		
15			11.03%	20.13%	32.39%	18.62%	5.03%
20				10.60%	26.34%	38.42%	35.11%
25					14.79%	32.78%	44.66%
30						23.09%	28.88%

Avg. Default bin
2.68% 44.66%

Average of Default bin broken down by Grade vs. Int Rate (bin). Colour shows average of Default bin. The marks are labelled by average of Default bin. The view is filtered on Exclusions (Grade,Int Rate (bin)), which keeps 25 members.

Loans associated with low FICO scores exhibit significantly higher default rates, confirming the strong predictive power of credit score in assessing borrower risk. However, rather than imposing a strict cutoff, the results suggest the need for risk-based pricing or tighter approval criteria for borrowers with FICO scores below approximately 650.

PD VIR & FICO

PD v IR & FICO

Last Fico Range High (bin)	Int Rate (bin) 5	10	15	20	25	30
450	79.43%	81.94%	84.37%	88.39%	91.95%	91.08%
500	66.82%	69.04%	73.17%	77.35%	81.27%	79.53%
550	48.74%	49.45%	55.74%	59.98%	63.65%	62.52%
600	19.29%	18.73%	23.76%	27.29%	25.70%	22.42%
650	1.95%	2.17%	2.94%	3.59%	2.76%	2.98%
700	0.40%	0.49%	0.71%	0.80%	0.95%	1.27%
750	0.22%	0.30%	0.33%	0.62%	0.65%	2.85%
800	0.29%	0.37%	0.29%	0.68%	0.71%	0.00%
850	1.75%	1.28%	7.14%	0.00%		0.00%

Average of Default binary broken down by Int Rate (bin) vs. Last Fico Range High (bin). Colour shows average of Default binary. The marks are labelled by average of Default binary. The data is filtered on Exclusions (Grade,Int Rate (bin)) and Exclusions (Int Rate (bin),Last Fico Range High). The Exclusions (Grade,Int Rate (bin)) filter keeps 25 members. The Exclusions (Int Rate (bin),Last Fico Range High) filter keeps 422 members.

Avg. Default binary

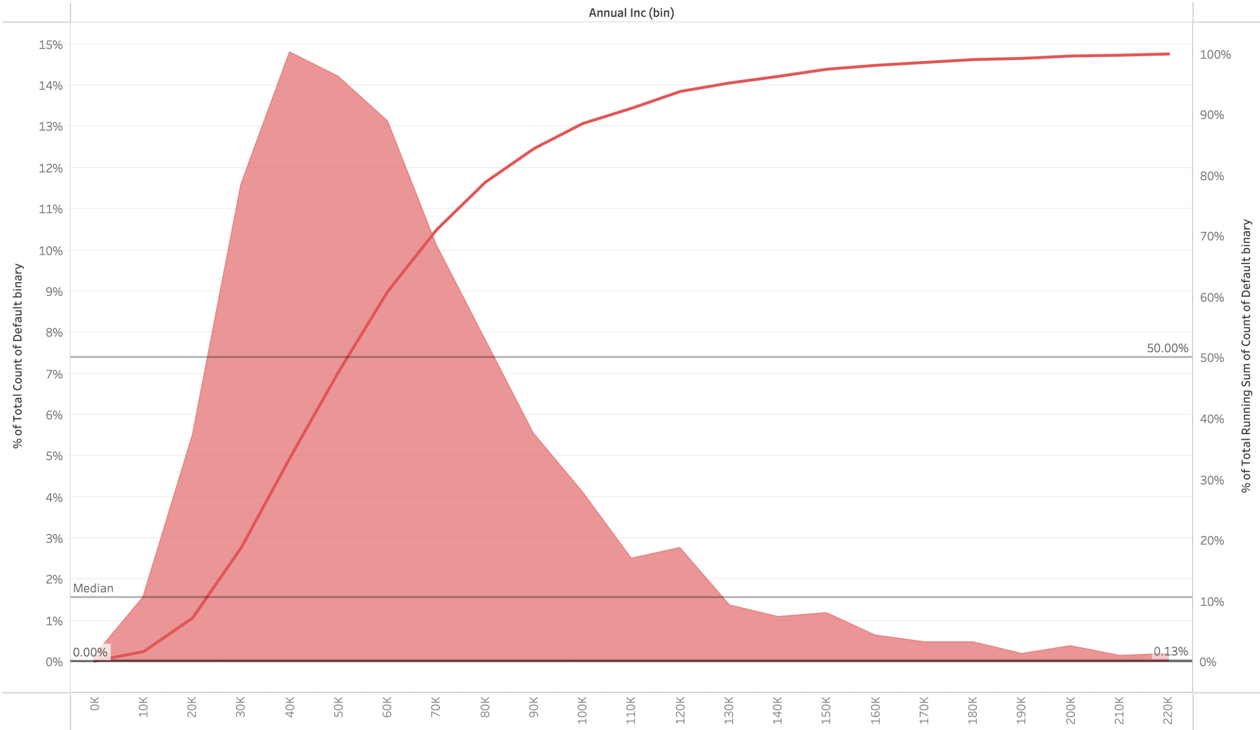
0.00% — 91.95%

PD v Inc

- 90% of all defaults are caused by lenders whose annual income is below the 100k mark.

- Median annual income for defaulters is 50K

%Default v Inc

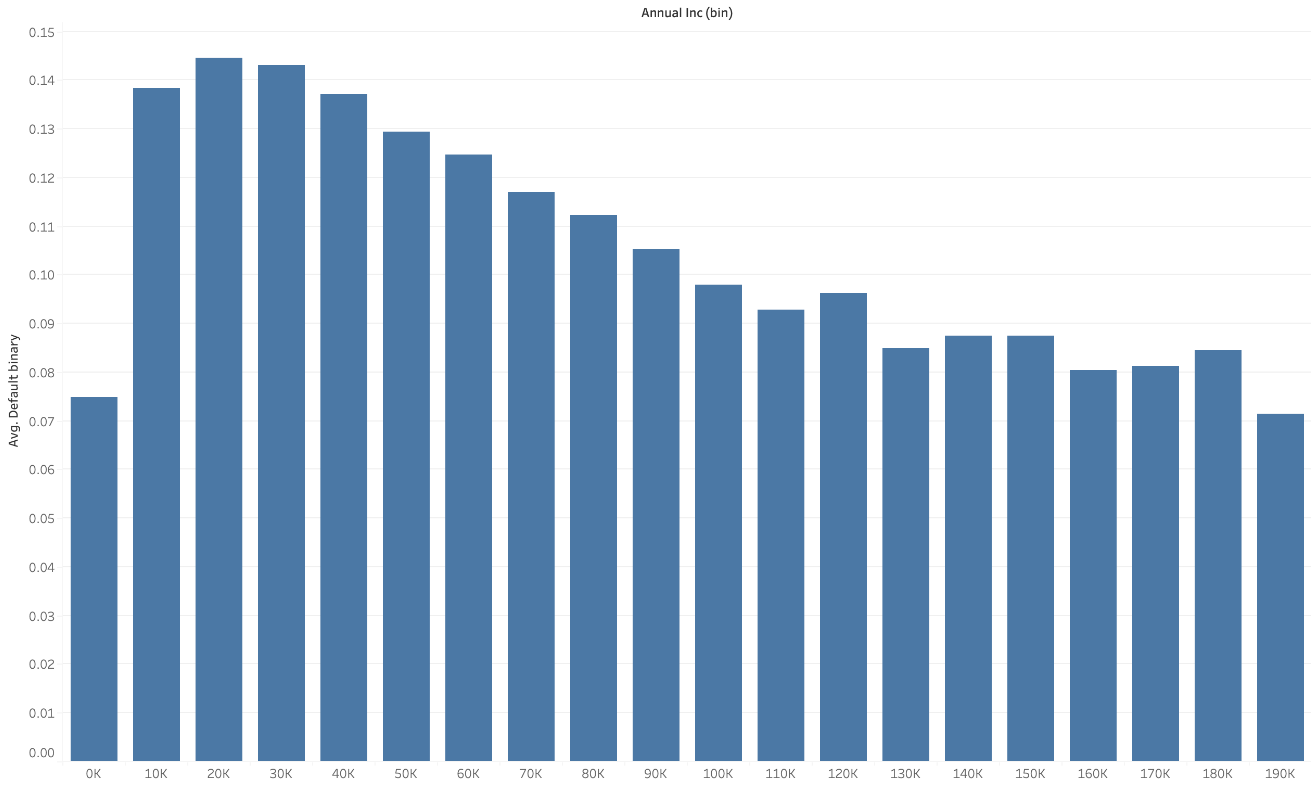


The trends of % of Total Count of Default binary along Table (Across then Down) and % of Total Running Sum of Count of Default binary along Table (Across) for Annual Inc (bin). Colour shows details about % of Total Count of Default binary along Table (Across then Down) and % of Total Running Sum of Count of Default binary along Table (Across). The data is filtered on Loan Status, which keeps Charged Off and Default. The view is filtered on Annual Inc (bin), which keeps 23 members.

Measure Names
- % of Total Count of Default binary along Table (Across then Down)
- % of Total Running Sum of Count of Default binary along Table (Across)

Annual income shows a negative relationship with default probability, but the magnitude of the effect is relatively limited compared to variables such as FICO score or interest rate. This suggests that income alone is not a strong predictor of default risk, and should be considered jointly with leverage indicators such as debt-to-income ratio.

PD v Inc



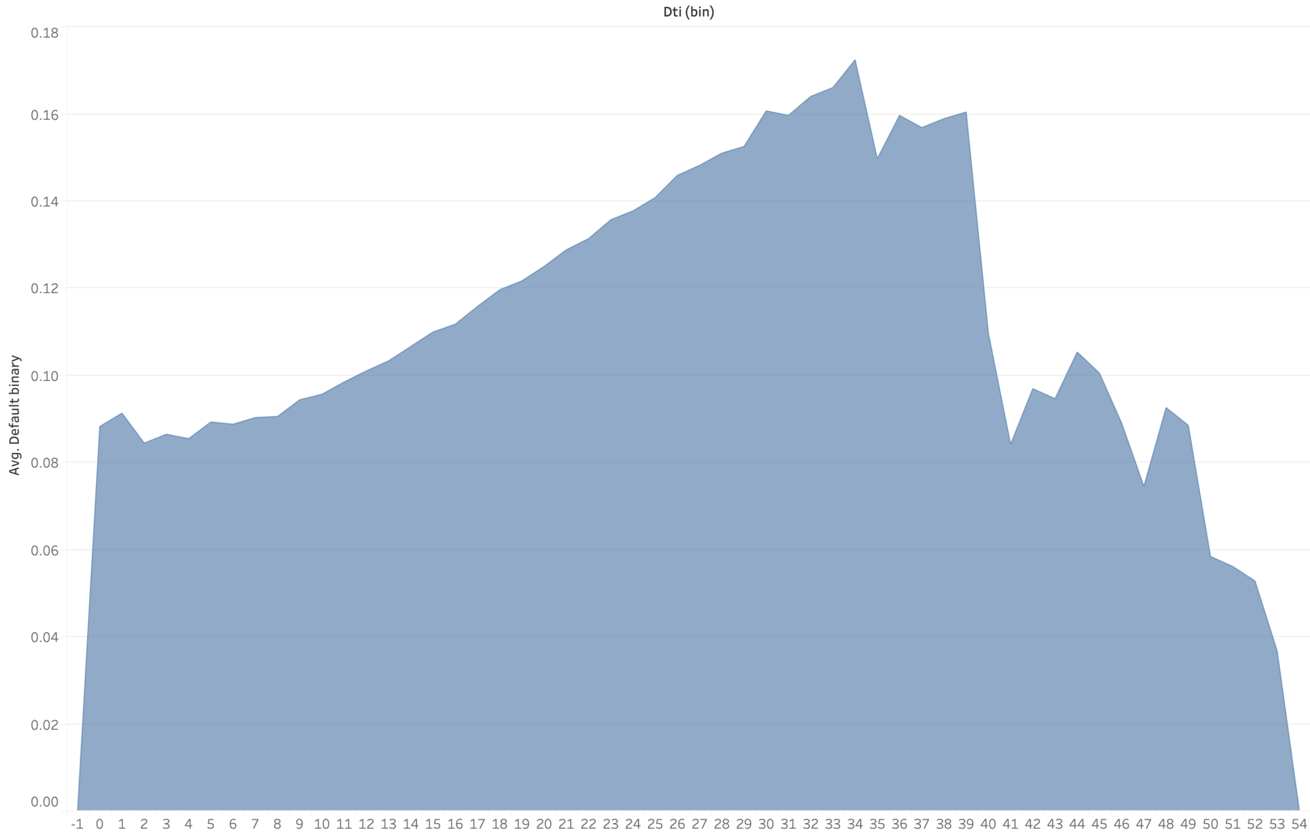
Average of Default binary for each Annual Inc (bin). The data is filtered on maximum of Annual Inc, which ranges from 9,999 to 200,000.

On Lending Club's website, DTI is defined as :

> "calculated using the borrower's total monthly debt payments on total debt obligations, excluding mortgage and the requested LendingClub loan, divided by the borrower's self-reported monthly income"

PD increases with DTI until a certain threshold, above which the bank seems to request other guarantees such as more stable employment. Also, the expenses a person has do not increase linearly with income, if a person doubles their income, their expenses do not increase two-fold. Therefore a person with high-salary high DTI might have more cushion to pay their debt than a person with low-salary low DTI as they need to spend less (in %), on expenses other than debt.

Sheet 5



Average of Default binary for each Dti (bin). The data is filtered on Dti, which ranges from -1 to 54.

The graph below shows that FICO score is a better measure at predicting PD.

PD v DTI & FICO

	Dti (bin)										
Last Fico Ra..	0	1	2	3	4	5	6	7	8	9	10
0	0.0000	0.0000	0.0000	0.0000	0.2000	0.0000	0.0000	0.3333	0.0000	0.1429	
480	0.7529	0.8144	0.8170	0.7688	0.7599	0.8057	0.7700	0.7961	0.7841	0.8055	0.9167
500	0.7471	0.6544	0.6577	0.6992	0.6619	0.7120	0.7003	0.7135	0.7404	0.7225	0.7500
520	0.6196	0.6214	0.5762	0.5777	0.6162	0.6050	0.6036	0.6350	0.6310	0.6426	0.7692
540	0.5823	0.5886	0.5102	0.5000	0.5556	0.5798	0.5436	0.5425	0.5670	0.5621	0.3846
560	0.4670	0.4365	0.4121	0.4333	0.4524	0.4473	0.4676	0.4564	0.4471	0.4964	0.4500
580	0.3426	0.3245	0.3419	0.3715	0.3148	0.3389	0.3729	0.3665	0.3593	0.3682	0.2500
600	0.2537	0.2220	0.2393	0.2052	0.2386	0.2307	0.2460	0.2567	0.2514	0.2562	0.2286
620	0.1567	0.1339	0.1196	0.1141	0.1159	0.1462	0.1382	0.1413	0.1483	0.1543	0.0741
640	0.0743	0.0513	0.0435	0.0482	0.0492	0.0554	0.0575	0.0544	0.0645	0.0612	0.0204
660	0.0332	0.0305	0.0177	0.0206	0.0209	0.0238	0.0218	0.0200	0.0209	0.0248	0.0135
680	0.0123	0.0187	0.0097	0.0104	0.0123	0.0081	0.0099	0.0113	0.0107	0.0107	0.0000
700	0.0162	0.0110	0.0071	0.0055	0.0078	0.0066	0.0061	0.0056	0.0056	0.0066	0.0088
720	0.0079	0.0083	0.0082	0.0047	0.0038	0.0041	0.0044	0.0045	0.0040	0.0040	0.0109
740	0.0120	0.0050	0.0025	0.0047	0.0035	0.0024	0.0027	0.0037	0.0029	0.0039	0.0000
760	0.0087	0.0062	0.0025	0.0035	0.0016	0.0017	0.0027	0.0025	0.0026	0.0025	0.0000
780	0.0185	0.0038	0.0032	0.0026	0.0059	0.0050	0.0027	0.0027	0.0025	0.0031	0.0204
800	0.0094	0.0066	0.0072	0.0061	0.0033	0.0055	0.0029	0.0043	0.0024	0.0018	0.0000
820	0.0174	0.0247	0.0034	0.0060	0.0000	0.0000	0.0052	0.0031	0.0014	0.0054	0.0000
840	0.0000	0.0667	0.0690	0.0625	0.0244	0.0000	0.0000	0.0182	0.0196	0.0370	

Average of Default binary broken down by Dti (bin) vs. Last Fico Range High (bin). Colour shows average of Default binary. The marks are labelled by average of Default binary. The data is filtered on Dti, which ranges from 0 to 10. The view is filtered on Dti (bin), which excludes -1.

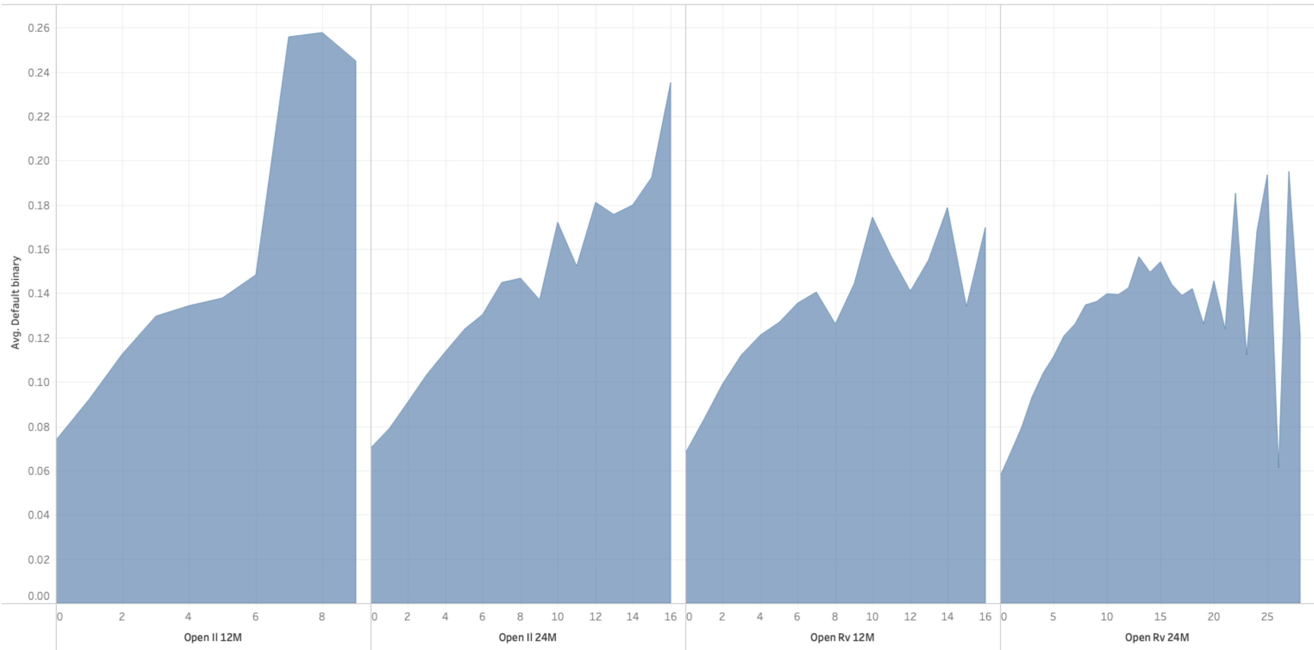
Avg. Default binary

0.0000 0.9167

PD v open accounts

PD increases with number of opened accounts, as individials who struggle to pay might open accounts one after the other in order to reimburse previous debt

PD v Open12M24M



The plots of average of Default binary for Open Il 12M, Open Il 24M, Open Rv 12M and Open Rv 24M. The data is filtered on sum of Count, which includes values greater than or equal to 30.

Variable	Impact on PD	Comment
FICO	High	Priced in IR but not used enough in loan decision making
Salary	Low	
Debt to Income	Moderate	Impactful until a certain threshold
Acc. open in last 12/24 months	Moderate	